

CSCE 650 VIRTUAL REALITY · MODULE 03 · LECTURE NOTES

Stereo Rendering

Two views, and the several ways to get them wrong.

MODULE OUTCOMES

- Construct asymmetric view frusta for an HMD.
- Explain why IPD must be the user's and not a default.
- Identify the standard stereo errors and their symptoms.
- Explain how UI and effects must be handled in stereo.
- Explain what the runtime supplies and what you must not override.

1 The geometry of stereo

DO NOT TOE IN THE CAMERAS

The intuitive construction places a camera at each eye and rotates both inward so their axes converge on a point of interest. **This is wrong**, and it is the mistake stereo photography made for most of a century before the geometry was properly understood.

Toeing in introduces vertical disparity. Rotating the two projections about different vertical axes means a point in the world projects to different *heights* in the two images, and the discrepancy grows toward the corners.

The visual system cannot fuse vertical disparity. It is equipped to handle horizontal disparity — that is what stereopsis is — and has no mechanism for vertical. The result is strain across the field and frank double vision near the edges.

The correct construction keeps the two view directions exactly parallel and makes the projection asymmetric instead. This is 'off-axis' or 'parallel asymmetric frustum' projection, it produces the same convergence, and it introduces no vertical disparity at all.

$$\text{eye}_{L,R} = \text{head} \pm (\text{IPD}/2) \cdot \text{right}$$

The two eye positions are offset along the head's right vector, and both look in the same direction. What differs is the **frustum**: the left, right, top, and bottom planes are not symmetric about the view axis, and all four differ between the eyes. **That asymmetry is the whole of stereo rendering**, and it is why a projection matrix cannot be built from a single field-of-view number as it can on a flat screen.

IPD IS A MEASUREMENT, NOT A CONSTANT

Interpupillary distance varies from roughly **52 to 78 mm** across the adult population, and is smaller in children. The mean is around 63 mm, which is the value hard-coded into a great deal of software that should not have hard-coded it.

Too wide and the world appears miniature. Greater separation produces greater disparity for the same scene, which the brain interprets as the scene being smaller and nearer — the user feels like a giant looking at a model.

Too narrow and the world appears gigantic, with distances stretched and everything feeling slightly unreal.

Users almost never identify the problem. They report that something feels off, that the space is strange, or simply that their eyes are tired. **Use the IPD the runtime reports**, which comes either from the headset's own measurement hardware or from the user's configured setting — and never substitute an average.

2 The standard errors

Error	Symptom from the inside
Toed-in cameras	Eye strain building over a minute or two; doubling of objects near the edges of the field. §1.
Wrong IPD	The world is the wrong size, in a way the user cannot name. Fatigue.
Swapped eyes	Depth is inverted — near objects appear far and vice versa, fighting every monocular cue. Deeply unpleasant and unmistakable , and worth inducing deliberately once, because it demonstrates how much work stereopsis is doing.
An effect rendered into one eye	Binocular rivalry (Module 02 §3) — the image alternates unstably.
UI at zero disparity	See below. The most common application-level stereo bug.
A view-dependent effect computed once and shared	Reflections and highlights sit at the wrong depth, so a mirror reads as a painting of a mirror.

UI MUST HAVE A DEPTH

A flat interface element composited identically into both eye buffers has **zero disparity**. The visual system reads zero disparity as *infinitely distant*.

But the element is drawn on top of everything, including objects a few centimetres from the face. So it is simultaneously the furthest thing in the scene by disparity and the nearest by occlusion.

These two cues are among the strongest the visual system has, and they are in direct contradiction.

The brain cannot resolve it. The result is eye strain and a diffuse discomfort that users consistently fail to attribute to the interface.

Put interface elements in the world, at a genuine depth — 1 to 2 m is comfortable and near the display's focal distance (Module 02) — and allow them to be occluded by nearer geometry. **Head-locked two-dimensional overlays are an instinct carried over from flat screens**, and they are one of the clearest markers of an application designed without VR in mind.

- **Compute per eye:** anything view-dependent — specular highlights, reflections, refraction, parallax-occlusion mapping — because each eye has its own view vector and the difference between them is real information the brain uses.
- **Compute per eye:** screen-space effects such as ambient occlusion and screen-space reflections. They are per-eye by construction, and they can disagree at silhouettes in ways that are visible; this is a known limitation rather than a bug to fix.
- **Orient billboards and impostors to the head**, not to either eye. Facing one eye's position makes them visibly disagree; facing the head makes them consistent and very slightly wrong for both, which is correct.
- **Share:** shadow map generation, lightmap baking, most culling, animation and skinning, and the majority of the shading setup. **This shareable fraction is exactly what stereo instancing and multiview exploit** (Module 10), and it is why two eyes cost considerably less than twice one.

3 Scale

SCALE ERRORS ARE INVISIBLE ON A MONITOR AND OBVIOUS IN VR

On a flat screen, a room modelled 20% too large simply reads as a room. There is no reference; the viewer has no body in the scene and no expectation to violate.

In VR the user has a body, and compares everything against it continuously and involuntarily. They know how tall they are and how far they can reach, and the comparison is automatic.

Doorways, stair risers, chair seats, table heights, and door handles are the giveaways — everyone has precise, unconsciously held expectations for these. Nobody has an expectation for the dimensions of a spaceship corridor, which is why invented environments tolerate scale error better than ordinary ones.

So build to real dimensions, in metres, and keep a known object — a correctly sized doorway, or a human figure — in the scene while working. **Assets authored for flat screens are frequently wrong**, because nobody had any reason to notice, and measuring them is a necessary part of porting anything to VR.

- **Place nothing closer than about 20 cm** — binocular fusion fails below that and the image doubles (Module 02).
- **Avoid large disparities at the edges of the field.** When the edge of the display cuts an object that the two eyes see differently, the conflict between the occluding frame and the stereo depth is uncomfortable. Stereo photography calls this a *window violation*, and the same geometry applies here.
- **Avoid rapid depth changes in whatever the user is attending to.** Vergence is a physical eye movement and takes time — forcing repeated large changes is tiring.
- **Keep the horizon level.** A tilted horizon conflicts directly with the vestibular system's own sense of down, and it is among the most reliably nauseating things you can do (Module 07).
- **Never move the horizon without the user's head moving.** The same rule, stated for the dynamic case, and it rules out a great many effects that are unremarkable on a flat screen.

TAKE THE MATRICES THE RUNTIME GIVES YOU

OpenXR reports, per frame and per eye: the eye pose, the four frustum half-angles, and the recommended render-target dimensions.

Those values account for the lens geometry, the display position, the user's configured IPD, and any per-device factory calibration. They cannot be reconstructed from a published field-of-view figure, and they differ between units of the same model.

Overriding them is almost always a mistake. It is the usual origin of applications with subtly incorrect stereo — the kind where users report fatigue and nobody can find a cause, because every individual component appears correct.

The legitimate exception is reducing the render target resolution for performance, which the runtime supports explicitly and which Module 10 covers. Scaling the resolution is fine; substituting your own projection is not.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 7 (free)	Visual rendering for VR, including the off-axis projection derivation of §1.
Stanford EE267 — stereo rendering lecture and assignment (free)	Builds the asymmetric frustum from scratch. Do the assignment; it makes the geometry concrete.
OpenXR specification — view configuration and projection (free)	What the runtime actually supplies, and in what coordinate conventions. The authority for §3's instruction.
Paul Bourke — calculating stereo pairs (free)	The clearest short explanation of why toe-in is wrong, with diagrams.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Construct asymmetric frusta from the runtime's reported angles and verify your projection matrices against the ones it supplies.
- 2 Deliberately implement toe-in stereo. Experience it, and measure the vertical disparity at the image corners.
- 3 Set the IPD 10 mm too wide and 10 mm too narrow. Write down what each does to apparent world scale.
- 4 **Swap the eye buffers** and experience inverted depth. This is worth doing exactly once.
- 5 Build a UI panel as a head-locked zero-disparity overlay, then as a world-space panel at 1.5 m. Compare the comfort over two minutes each.
- 6 Render a reflective sphere with the reflection computed from one eye and shared. Describe where the reflection appears to sit.
- 7 Build a room to real measured dimensions, then rebuild it 20% larger. Have someone else judge which is correct without being told.
- 8 Place a door, a chair, and a staircase at correct dimensions and deliberately wrong ones, and record which error is noticed fastest.
- 9 Tilt the horizon by 5 degrees and experience it for thirty seconds. Stop when uncomfortable.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is toe-in wrong, and what does it introduce that the eyes cannot handle?
- 2 Describe the correct stereo construction in one sentence.
- 3 What is the range of human IPD, and what do wrong values do?
- 4 Give six stereo errors and the symptom of each.
- 5 Why does zero-disparity UI cause strain, and what is the fix?
- 6 What must be computed per eye, and what can be shared?
- 7 Why are scale errors obvious in VR and invisible on a monitor?
- 8 Give five stereo comfort rules.
- 9 What does the runtime supply, and what is the one legitimate thing to override?